

Name: _____

Date: _____

Class: _____

MyDesign

Portfolio

Element G Initial Template

How to use this template

Teachers: You may customize this document before providing it to students.

Students: The template below is one way to present your construction of a testable prototype, but the specific assignment given by your instructor may require a different structure or dictate different details. If you make a copy of this template to edit for your prototype construction details, be sure to delete the instructions.

Guidance before filling out the template that follows:

- Because you will iterate on your design multiple times, you will need to create multiple copies of these sections. Think of this Element as another piece of a diary that you will date and keep adding to throughout your work until your final iteration. Iteration must shine through in this Element also.
- You will then proceed to test and analyze this design and then determine your next iteration, returning to this Element afterward when you are ready to construct a new testable prototype. Fill out each of these steps again.

Element G: Construction of a Testable Prototype

Final Prototype Iteration Explanation (G1)

This section focuses on the process of explaining your prototypes. While the title of the subelement focuses on your final prototype, you must have at least an initial prototype and a final prototype, with additional prototypes in between being desirable. Note that you never know which prototype is your final prototype until after you test it, so be sure to record all of the information for each prototype. Use as many copies of this sub-template as needed.

Name: _____

Date: _____

Class: _____

Ensure that photos are taken from multiple angles so that all sides and subsystems can be seen. All images should have captions and should be referenced in-text.

Prototype Construction of Design #1 (make as many copies, updating the design number, as needed)

Date:

Photos with captions of your full construction process from day 1 of building through to the last day of constructing your final prototype iteration:

Details about all steps of construction (add step numbers as needed), including the tools and manufacturing methods used:

	Details	Tools and Manufacturing Methods Used
Step 1	Measure where the holes need to be and mark the correct spots.	Ruler, sharpie, storage container
Step 2	Cut a hole in the lid for the fan and air filter	Dremel, safety goggles, storage container
Step 3	Make a hole in the side of the storage bin for the cord	Dremel, safety goggles, storage container
Step 4	Carve vents on the long sides of the storage bin	Dremel, safety goggles, storage container
Step 5	Sand down all of the sharp edges.	Sanding paper
Step 6	Place the fan and the air filter inside	Air filter, fan, storage container

Name: _____

Date: _____

Class: _____

	the storage container.	
Step 7	Add barriers inside the box to control airflow and ensure that all of the air has to go through the filter.	Cardboard, storage container, cord covers
Step 8	Test if the purifier works	Completed purifier, air quality meter

Notes of the progression from your blueprint to your (various) physical prototype(s), including why and how aspects of your design may have changed:

Final Prototype Construction of Design #2 (change this number to match the design number you're on)

Date:

Photos with captions of your full construction process from day 1 of building through to the last day of constructing your final prototype iteration:

Details about all steps of construction (add step numbers as needed), including the tools and manufacturing methods used:

	Details	Tools and Manufacturing Methods Used
Step 1		
Step 2		
Step 3		

Name: _____

Date: _____

Class: _____

Step 4		
Step 5		
Step 6		
Step 7		

Notes of the progression from your blueprint to your (various) physical prototype(s), including why and how aspects of your design may have changed:

List of material details, including item name, description, where it was purchased/obtained (include URL if applicable), cost per unit (if applicable), # of units, total cost (if applicable)

Material name	Description	Where material was purchased	Cost per unit	# units	Total cost per item
1 Overhead Storage Bin Rack		Amazon	\$23.99	1	\$23.99
Cord Hider Wall Cable Cover 142in		Amazon	\$9.99	1	\$9.99
10 ft Extension Cord		Amazon	\$10.99	1	\$10.99
HDX 27 gal. Strong Box Plastic Storage Tote		Amazon	\$29.40	1	\$29.40

Name: _____

Date: _____

Class: _____

Amazon Basic 3 Speed Box Fan 20-inch		Amazon	\$41.39	1	\$41.39
Accumulair Carbon 20x20x1 Air Filter		Amazon	\$16.05	1	\$16.05

Detailed sketches, which might include CAD if you have learned it in your class:

Construction of the Final Prototype (G2)

When you've finished construction of your final prototype, you must be sure that your prototype allows sufficient functionality that can collect objective, measurable data for all or nearly all of your design requirements. This requires you to look back at your design requirements from Element C. Copy all of your design requirements from Element C into this table and then add how your prototype will allow you to collect objective, measurable data. Note any weaknesses you find or details that will require expert review because they cannot be mathematically modeled or tested,

For example, if you have a design requirement that your design can "transport medicine a distance of 5 meters in 1 minute", is your prototype sufficiently functional such that you can test that? (e.g. can the prototype move the medicine at least a little bit?)

Design Requirement	How will this prototype allow you to collect objective, measurable data?
Size- 16"x16"	Yes, we can measure it to ensure that it fits within the criteria.
Fan	Yes, the fan will bring air into and push air out of the purifier.
Chargeable	No
Long lasting	Yes, we can record how long the filter can be used before it needs to be replaced.

Name: _____

Date: _____

Class: _____

Easy to manage	No
Quiet	Yes, we can listen to see if it causes significant noise pollution.
Diffuser to release essential oils	No

Testing or Modeling of Attributes (G3, G4) and Addressing Non-Testable Attributes (G4)

Describe all of your prototype's attributes (features, aspects, details, or subsystems). Demonstrate that your prototype is constructed with sufficient functionality so that your prototype can test all attributes of your design. This requires you to look back at your initial proposed design details from Element D.

For example, if your initial design details called for wheels and motors, is your prototype sufficiently functional such that you can test out those wheels and motors?

If there are aspects of your design that you believe cannot be physically prototyped or mathematically modeled within the scope and limits of your class, you must later provide evidence that an expert has reviewed your justifications and that you have incorporated their feedback. You do not need to conduct that review in Element G. For now, enter "expert review required" for these attributes in the final column.

Attribute	Description	Describe your prototype's functionality to test this attribute
Vents (holes)	It's the ability to take in and push out air (airflow)	The box fan is in the middle with vents on each side. On one side the vents act as an intake bringing the bad air

Name: _____

Date: _____

Class: _____

		into the box, and it goes through the fan & filter leaving the good air going outwards through the opposing side vents.
Lid	The lid can hold our air filter (fan and filter)	The lid traps air in our box and makes it flow down and out into the filter.
Ability to filter out Particulate Matter	The air filter can trap the particulate matter before the air leaves the purifier.	When the air is brought into the purifier and pushed through the filter, the filter can trap the particulate matter, it is then tested with an air quality meter.
Durability	Its ability to undergo wear and tear throughout time.	The box acts as a protection for the filter and the fan to make sure it works to its best ability.
Ability to filter out Volatile Organic Compounds	The air filter is able to take out the Volatile Organic Compounds from the air.	After the polluted air comes in contact with the filter, the volatile organic compounds are left in the filter before the clean air is then pushed out and tested with an air quality meter.
Ability to filter out Formaldehyde particles	The air filter can stop the Formaldehyde particles from going through the purifier.	The fan brings impure air into the purifier, where the Formaldehyde particles are then trapped before the, now pure, air is released. We will then test it with an air quality meter.



Name: _____

Date: _____

Class: _____

Safe	It can take out all of the particles in the air that are harmful to us.	Fan blades are protected and the box is cleaned up without any loose parts. Our cord also bundles up .
-------------	---	--

Addressing Non-Testable Attributes (G4)

For all attributes that you listed "expert review required" in the table above, justify why that is true. "Field experts" are professionals who spend at least a portion of their time working in a field directly related to your project. These field experts are not permitted to double count as stakeholders.

Attribute	Justification for why this attribute cannot be modeled or tested with your prototype
N/A	

Original Measurements:

	Original Measurements	Fire Measurements	Air Filter Measurements
PPM 2.5 (particles per minute)	609	026	026
PM 1	299	019	
PM 10	829	030	
Time (only for	0 mins 0 sec		3 mins and 58 sec



Name: _____

Date: _____

Class: _____

air filter)			
-------------	--	--	--

